Differential Equations Workbook

Systems of differential equations

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MATH

MATRIX BASICS

■ 1. Find $3A - (1/2)B + 2C$.

$$A = \begin{bmatrix} 0 & 1 & -\frac{1}{3} \\ 2 & \frac{1}{3} & 4 \\ \frac{2}{3} & -1 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 2 & 0 & -4 \\ -2 & 6 & 4 \\ 10 & 0 & 0 \end{bmatrix} \qquad C = \begin{bmatrix} 1 & 0 & 3 \\ -2 & 10 & 1 \\ -4 & 3 & 0 \end{bmatrix}$$

■ 2. For the 2×2 matrix, compute the determinant.

$$\begin{bmatrix} \frac{3}{2} & 1 \\ 2 & -6 \end{bmatrix}$$

■ 3. For the 3×3 matrix, compute the determinant.

$$\begin{bmatrix} -1 & 5 & 0 \\ \frac{1}{2} & -\frac{3}{2} & 1 \\ 4 & 0 & 2 \end{bmatrix}$$

■ 4. Find the Eigenvalues and Eigenvectors of the matrix.

$$A = \begin{bmatrix} 1 & 3 \\ -1 & 5 \end{bmatrix}$$

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■ 5. Find the Eigenvalues and Eigenvectors of the matrix.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ -4 & -4 & 0 \\ -2 & 1 & 2 \end{bmatrix}$$

■ 6. Find the Eigenvalues and Eigenvectors of the matrix.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 0 & -9 \\ 4 & 1 & 0 \end{bmatrix}$$

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BUILDING SYSTEMS

■ 1. Rewrite the system of differential equations in matrix form.

$$x_1' = 3x_1 - 4x_3 + e^t$$

$$x_2' = 5x_2 + x_1 - 13x_3$$

$$x_3' = \sin(t^2) - 3x_1 - 2x_2$$

■ 2. Rewrite the system of differential equations with the initial conditions in matrix form.

$$x_1' = -x_1 + 2x_3 - t^2 \qquad x_1(0) = 2$$

$$x_2' = 9x_2 - x_3 \qquad x_2(0) = -3$$

$$x_3' = 2x_2 - 4x_3 + 5\cos t \qquad x_3(0) = 0$$

■ 3. Convert the fourth order linear differential equation into a system of differential equations in matrix form.

$$3y^{(4)} + 6y''' - 18y'' - 20t\cos(t^2) = e^t$$

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■ 4. Convert the given initial value problem into a system of differential equations and rewrite it as a matrix equation.

$$y''' - 4ty' + 7y - t^2y'' = 2t^3 - 4$$

$$y(0) = -2,\ y'(0) = 1,\ y''(0) = 0 \qquad \vec{x}(0) = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

■ 5. Convert the third order linear differential equation into a system of differential equations in matrix form.

$$e^ty''' - 5y'' + y' - t^2y + e^{2t} = 0$$

■ 6. Convert the fourth order linear differential equation with the given initial conditions into a system of differential equations in matrix form.

$$y^{(4)} - (\cos t)y''' + 3e^ty'' - 2y = 0$$

$$y(0) = 3,\ y'(0) = 1,\ y''(0) = -2,\ y'''(0) = 0$$

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SOLVING SYSTEMS

■ 1. Show that the vector is a solution to the system.

$$x_1' = x_1 + 3x_2 - e^{2t}$$

$$x_2' = 3x_1 + x_2$$

$$\overrightarrow{x_1} = \begin{bmatrix} -3 \\ 3 \end{bmatrix} e^{-2t} + \begin{bmatrix} \frac{1}{8} \\ \frac{3}{8} \end{bmatrix} e^{2t}$$

■ 2. Show that the vector is a solution to the system.

$$\overrightarrow{x}' = \begin{bmatrix} 2 & 1 & 1 \\ -2 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} \overrightarrow{x}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} e^{t} + \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} e^{2t}$$

■ 3. Confirm that the vectors are linearly independent.

$$\overrightarrow{x_1} = \begin{bmatrix} 3 \\ -3 \\ 7 \end{bmatrix} e^{3t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{6t} \qquad \overrightarrow{x_3} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{7t}$$

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■ 4. Verify that the solution vectors satisfy the system of differential equations. Calculate the Wronskian of the solution vectors and write the general solution of the homogeneous system of differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \vec{x}$$

$$\{\vec{x_1}, \vec{x_2}, \vec{x_3}\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{t}, \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} e^{-t}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} e^{2t} \right\}$$

■ 5. Verify that the solution vectors satisfy the system of differential equations. Calculate the Wronskian of the solution vectors and write the general solution of the homogeneous system of differential equations.

$$\vec{x}' = \begin{bmatrix} 6 & 5 \\ -5 & 0 \end{bmatrix} \vec{x}$$

$$\{\vec{x_1}, \vec{x_2}\} = \left\{ \begin{bmatrix} -3\cos(4t) + 4\sin(4t) \\ 5\cos(4t) \end{bmatrix} e^{3t}, \begin{bmatrix} -4\cos(4t) - 3\sin(4t) \\ 5\sin(4t) \end{bmatrix} e^{3t} \right\}$$

■ 6. Verify that the solution vectors satisfy the system of differential equations. Calculate the Wronskian of the solution vectors and write the general solution of the homogeneous system of differential equations.

$$\vec{x}' = \begin{bmatrix} 5 & -1 & -1 \\ 0 & 4 & -1 \\ 0 & -1 & 4 \end{bmatrix} \vec{x}$$

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$$\{\vec{x_1}, \vec{x_2}, \vec{x_3}\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{3t}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{5t}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} e^{5t} \right\}$$

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DISTINCT REAL EIGENVALUES

■ 1. Solve the system of differential equations.

$$x_1' = 4x_1 + 8x_2$$

$$x_2' = x_1 + 2x_2$$

■ 2. Solve the system of differential equations.

$$x_1' = 7x_1 + 6x_2 + 3x_3$$

$$x_2' = 2x_2 - x_1 + x_3$$

$$x_3' = x_1 + 2x_2 + x_3$$

■ 3. Solve the system of differential equations, given $x_1(0) = -2$ and $x_2(0) = 19$.

$$x_1' = x_1 + 2x_2$$

$$x_2' = 5x_1 - 2x_2$$

■ 4. Convert the differential equation into a system of equations, then use the system to solve the initial value problem, given $y(0) = 2$ and $y'(0) = 6$.

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$$y'' - 6y' - 7y = 0$$

■ 5. Convert the differential equation into a system of equations, then solve the system.

$$y''' - 2y'' - 9y' + 18y = 0$$

■ 6. Solve the initial value problem, given $y(0) = 2$, $y'(0) = 6$, and $y''(0) = 16$.

$$y''' - 6y'' + 11y' - 6y = 0$$

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EQUAL REAL EIGENVALUES WITH MULTIPLICITY TWO

■ 1. Solve the system of the differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} \vec{x}$$

■ 2. Solve the system of the differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix} \vec{x}$$

■ 3. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 3 & -1 & 2 \\ 2 & 0 & 4 \\ -1 & 1 & 0 \end{bmatrix} \vec{x}$$

■ 4. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \vec{x}$$

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■ 5. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 5 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 5 \end{bmatrix} \vec{x}$$

■ 6. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & 0 & 4 \\ 0 & -3 & 0 \\ 4 & 0 & 1 \end{bmatrix} \vec{x}$$

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EQUAL REAL EIGENVALUES WITH MULTIPLICITY THREE

■ 1. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \vec{x}$$

■ 2. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \vec{x}$$

■ 3. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \vec{x}$$

■ 4. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 0 & 2 & 6 \\ -2 & -4 & -5 \\ 0 & 0 & -2 \end{bmatrix} \vec{x}$$

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■ 5. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} -3 & 2 & 0 \\ -2 & 1 & 0 \\ -2 & 2 & -1 \end{bmatrix} \vec{x}$$

■ 6. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \vec{x}$$

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COMPLEX EIGENVALUES

■ 1. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 2 & 4 \\ -2 & -2 \end{bmatrix} \vec{x}$$

■ 2. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} -6 & 4 \\ -8 & 2 \end{bmatrix} \vec{x}$$

■ 3. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & -1 \\ 5 & 5 \end{bmatrix} \vec{x}$$

■ 4. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 6 & -10 \\ 5 & -4 \end{bmatrix} \vec{x}$$

■ 5. Find the general solution to the system of differential equations.

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$$\vec{x}' = \begin{bmatrix} -2 & 0 & 0 \\ 1 & 2 & 5 \\ 2 & -1 & 4 \end{bmatrix} \vec{x}$$

6. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 2 \\ 0 & -5 & 1 \end{bmatrix} \vec{x}$$

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PHASE PORTRAITS FOR DISTINCT REAL EIGENVALUES

■ 1. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} -4 & 4 \\ -2 & 5 \end{bmatrix}$$

■ 2. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 2 & 1 \\ 4 & -1 \end{bmatrix}$$

■ 3. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$$

■ 4. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$$

■ 5. Sketch the phase portrait of the system.

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$$A = \begin{bmatrix} -2 & 4 \\ 0 & 0 \end{bmatrix}$$

6. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} -3 & -1 \\ 0 & -4 \end{bmatrix}$$

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PHASE PORTRAITS FOR EQUAL REAL EIGENVALUES

1. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 6 & 2 \\ -8 & -2 \end{bmatrix}$$

2. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} -3 & -4 \\ 4 & 5 \end{bmatrix}$$

3. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} -1 & 9 \\ -1 & -7 \end{bmatrix}$$

4. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 10 & 4 \\ -4 & 2 \end{bmatrix}$$

5. Sketch the phase portrait of the system.

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$$A = \begin{bmatrix} 16 & -5 \\ 5 & 6 \end{bmatrix}$$

6. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix}$$

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PHASE PORTRAITS FOR COMPLEX EIGENVALUES

1. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 5 & 1 \\ -5 & 2 \end{bmatrix}$$

2. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 1 & -3 \\ 2 & 2 \end{bmatrix}$$

3. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 2 & -1 \\ 10 & -4 \end{bmatrix}$$

4. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} -2 & 2 \\ -1 & 0 \end{bmatrix}$$

5. Sketch the phase portrait of the system.

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$$A = \begin{bmatrix} -2 & -4 \\ 2 & 2 \end{bmatrix}$$

6. Sketch the phase portrait of the system.

$$A = \begin{bmatrix} 1 & 2 \\ -13 & -1 \end{bmatrix}$$

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UNDETERMINED COEFFICIENTS FOR NONHOMOGENEOUS SYSTEMS

■ 1. Use undetermined coefficients to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} \cos t \\ 2 \sin t \end{bmatrix}$$

■ 2. Use undetermined coefficients to find the general solution to the initial value problem.

$$\vec{x}' = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} 2e^{3t} \\ t^2 \end{bmatrix}$$

$$\vec{x}(0) = \begin{bmatrix} \frac{1}{4} \\ \frac{3}{4} \end{bmatrix}$$

■ 3. Use undetermined coefficients to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ 2 & -1 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} 6e^{4t} \\ \cos(3t) \\ 1 \end{bmatrix}$$

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■ 4. Use undetermined coefficients to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix} \vec{x} + \begin{bmatrix} -e^t \\ 3t^2 \\ 2e^{-t} \end{bmatrix}$$

■ 5. Use the method of undetermined coefficients to find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix} \vec{x} + \begin{bmatrix} t + e^t \\ 2t^2 + 1 \end{bmatrix}$$

■ 6. Use undetermined coefficients to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 \\ -4 & 4 & 0 \\ -2 & 1 & 2 \end{bmatrix} \vec{x} + \begin{bmatrix} 3e^t \\ 3t + 2e^{3t} \\ t^3 - 4e^t \end{bmatrix}$$

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VARIATION OF PARAMETERS FOR NONHOMOGENEOUS SYSTEMS

■ 1. Use the method of variation of parameters to find the general solution to the system, given the complementary solution.

$$\vec{x}' = A\vec{x} + \begin{bmatrix} e^{-3t} \\ 2e^{t} + 1 \end{bmatrix}$$

$$\vec{x_c} = c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 2 \\ 7 \end{bmatrix} e^{5t}$$

■ 2. Find the general solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 6 & -8 \\ 4 & -6 \end{bmatrix} \vec{x} + \begin{bmatrix} 2t + 1 \\ e^{t} \end{bmatrix}$$

■ 3. Solve the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} \sin t + \cos t \\ \sin t \end{bmatrix}$$

■ 4. Solve the system of differential equations.

$$x_1' = -5x_1 + 12x_2 + e^{t}$$

$$x_2' = -4x_1 + 9x_2 + 3e^{-t}$$

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■ 5. Solve the system of differential equations.

$$x_1' = x_1 + 3x_2 + 1 + e^{2t}$$

$$x_2' = x_2 + e^t$$

■ 6. Solve the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} t^5 \\ t^3 \end{bmatrix}$$

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THE MATRIX EXPONENTIAL

■ 1. Use an inverse Laplace transform to calculate the matrix exponential.

$$A = \begin{bmatrix} 1 & -2 \\ 3 & -1 \end{bmatrix}$$

■ 2. Use the matrix exponential to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 1 & -1 \\ -3 & 3 \end{bmatrix} \vec{x}$$

■ 3. Use the matrix exponential to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 0 & -1 \\ -3 & 2 \end{bmatrix} \vec{x}$$

■ 4. Use the matrix exponential to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} 6t^2 - te^{-3t} \\ \sin t \end{bmatrix}$$

■ 5. Use the matrix exponential to find the general solution to the system.

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$$\vec{x}' = \begin{bmatrix} -1 & 0 & 0 \\ -2 & 0 & 3 \\ 3 & -1 & -4 \end{bmatrix} \vec{x}$$

6. Use the matrix exponential to find the general solution to the system.

$$\vec{x}' = \begin{bmatrix} 0 & 1 & 4 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ e^{3t} \\ 6t^2 \end{bmatrix}$$

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